Module 07: Lab 07

Nischal Pant

1. **What is the main difference between unsupervised and supervised learning?**

The main difference between unsupervised and supervised learning is the labeled data.

In supervised learning, the algorithm is trained on a dataset that includes input-output pairs, the data uses different variables to predict a certain variable.

Unsupervised learning uses data with no labels to find patterns, structures, and/or relationships within the data without any output variable.

2. **Is centering required for PCA? Is scaling required for PCA? Explain your answers.**

Yes, centering is required for PCA to ensure that the data has a mean of zero, which is necessary for the covariance matrix to be correctly computed. Failure to do so results in the principal components not being represented in the directions of maximum variance accurately.

Scaling is not required for PCA but should be done. Features with different units or scales might not equally contribute to the analysis, causing features with larger scales to have a greater influence on the principal components.

3. **After running the code above, run the following code to obtain the PCA feature loadings. Copy and paste the output. Which feature has the strongest influence on the first PC? Which feature has the weakest influence on the first PC?**

```
```{r}
pca_results$rotation
```
```

```
              PC1         PC2         PC3          PC4          PC5
 [1,]  0.12274182  0.36061844 -0.34398989 -0.065639783 -0.601832136
 [2,]  0.48802274 -0.15045681  0.09014459 -0.076896967 -0.036599404
 [3,] -0.09747898 -0.52567045 -0.14365080 -0.381939447 -0.146273632
 [4,]  0.12102885  0.02357670 -0.64654694  0.226369582  0.163150216
 [5,]  0.10009491 -0.21646946 -0.61265540  0.136531963  0.190085913
 [6,]  0.11271156  0.39371822 -0.13344151 -0.529490728 -0.276353240
 [7,] -0.45140570  0.34994847 -0.03440842  0.293595917  0.007372956
 [8,]  0.38600817  0.35130174  0.13091172  0.358564394  0.127084993
 [9,]  0.58800451 -0.02092786  0.11759108  0.002548485  0.083478225
[10,] -0.03710146  0.34963398 -0.09789614 -0.528093227  0.672879824
               PC6         PC7          PC8          PC9          PC10
 [1,] -0.576461323  0.05112682 -0.162037304  0.094130848 -3.305778e-03
 [2,] -0.032960233  0.65150985  0.482254145  0.256313803  7.484096e-04
 [3,] -0.006828669  0.38085801 -0.374041053 -0.473503929  1.305643e-01
 [4,]  0.008709919 -0.09147813  0.455900447 -0.522131265 -1.284155e-03
 [5,]  0.237582625  0.04943666 -0.368520524  0.564185114  3.600662e-05
 [6,]  0.669537280 -0.06264737  0.070821490 -0.022791326  1.973253e-03
 [7,]  0.139814283  0.46059149  0.002935337  0.006100992  5.954777e-01
 [8,]  0.158830961  0.30527322 -0.469025342 -0.321100442 -3.524466e-01
 [9,] -0.042368718 -0.30498204 -0.167257449 -0.078129820  7.100151e-01
[10,] -0.339392667  0.12157680 -0.068054558  0.023330608 -1.466239e-03
```

According to our output, the strongest influence on the first principle component is by Feature 9 (0.58800451) and the weakest influence is by Feature 10 (-0.03710146)
.